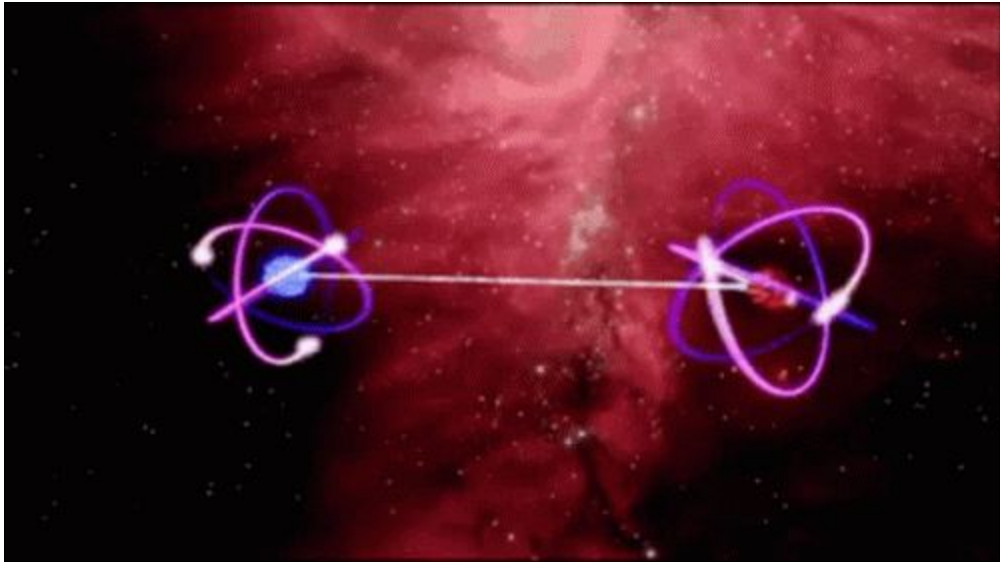
Significant-Loophole-Free Test of Bell’s Theorem with Entangled Photons

A presentation of Giustina et al 2015

David Brin, Nikolaj Topsøe, Þorri Elís Halldóruson, Janusz Wilczek

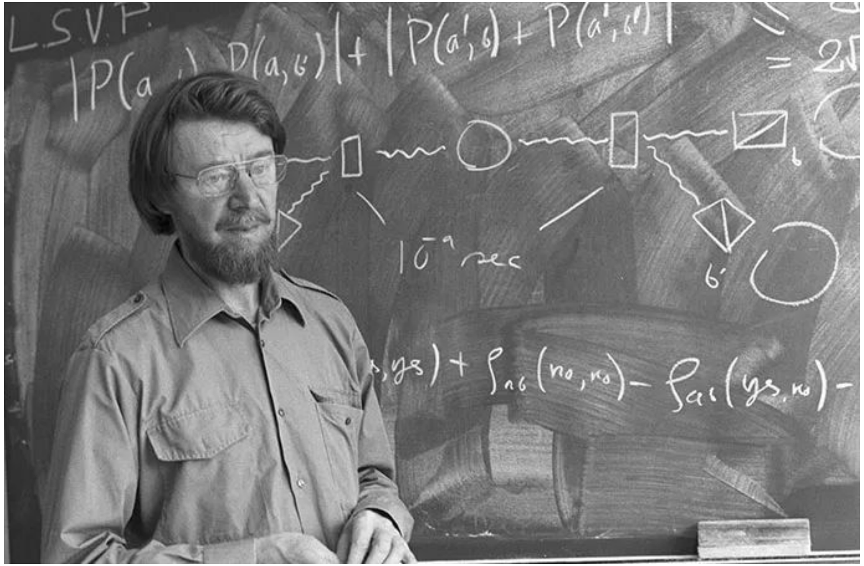
Overview

- Bell’s theorem
- Loopholes closed
- Results



Overview

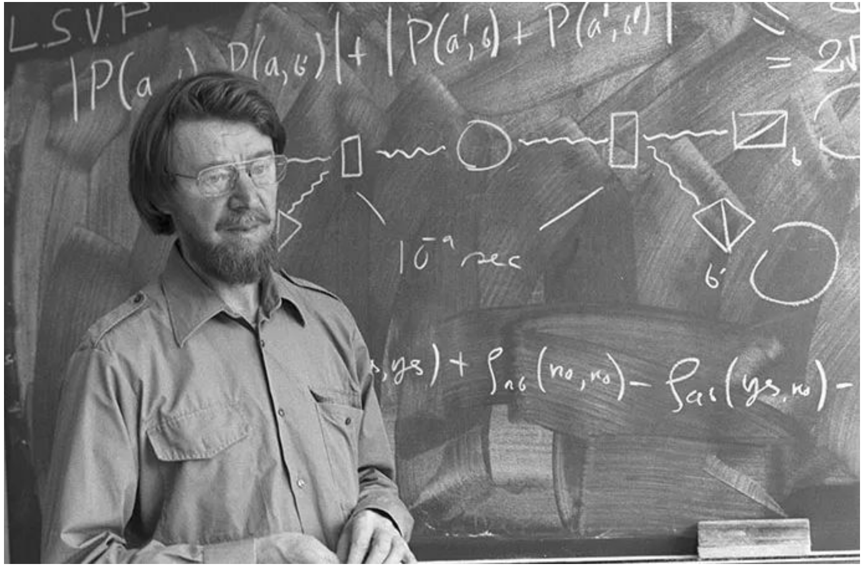
- **Bell's theorem**
- Loopholes closed
- Results



$$|S| = |\,\langle A_1, B_1\rangle + \langle A_1, B_2\rangle + \langle A_2, B_1\rangle - \langle A_2, B_2\rangle\,| \leq 2$$

Overview

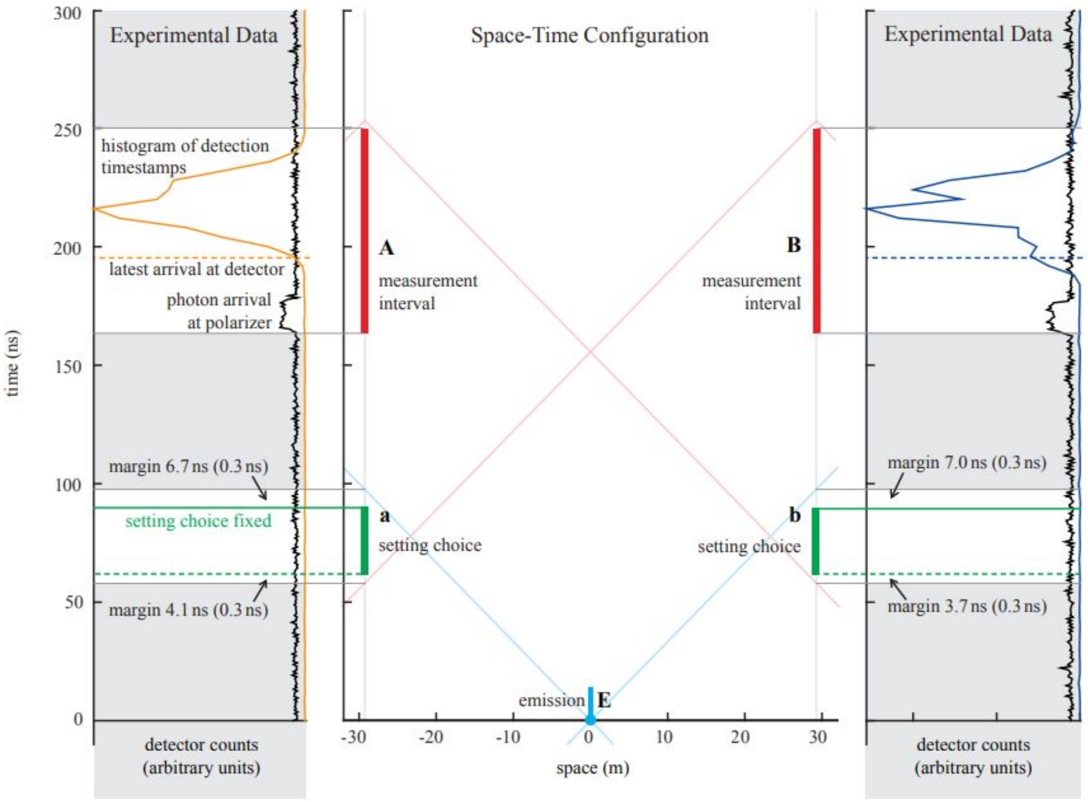
- **Bell's theorem**
- Loopholes closed
- Results



$$J \equiv p_{++}(a_1, b_1) - p_{+0}(a_1, b_2) - p_{0+}(a_2, b_1) - p_{++}(a_2, b_2) \leq 0$$

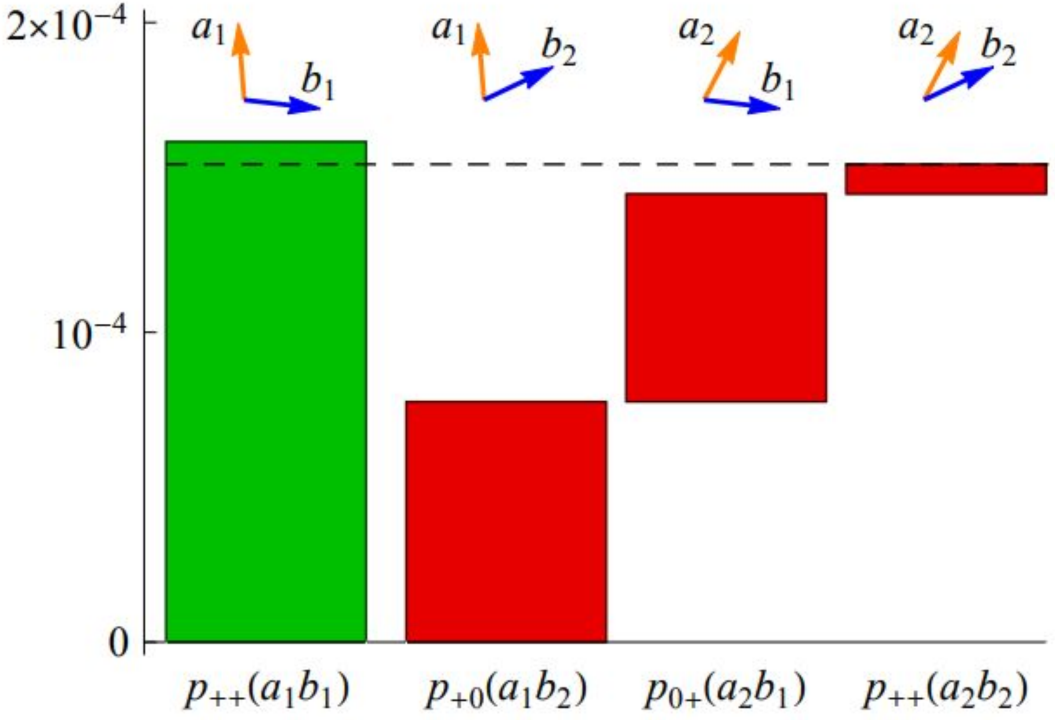
Overview

- Bell’s theorem
- **Loopholes closed**
- Results

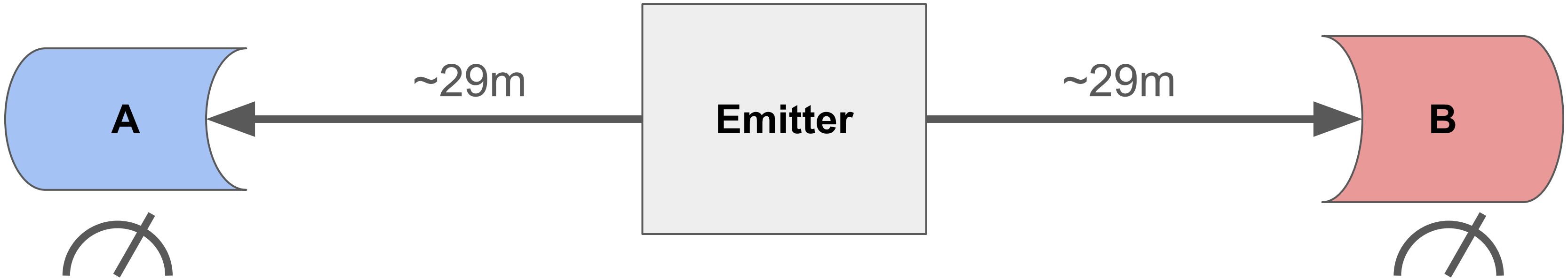


Overview

- Bell's theorem
- Loopholes closed
- **Results**



Experimental setup



- Pairs of entangled photons emitted to A and B.
- Rapid choice of measurement setting at both sites.
- For the randomized settings a1/a2 and b1/b2 we record the outcomes.

Experimental setup

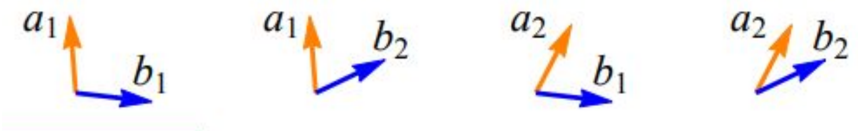
$$J \equiv p_{++}(a_1, b_1) - p_{+0}(a_1, b_2) - p_{0+}(a_2, b_1) - p_{++}(a_2, b_2) \leq 0$$

- Estimate the four terms by detection frequencies: $\hat{p}_{++}(a_1, b_1) = \frac{n_{(++|a_1,b_1)}}{N_{(a_1,b_1)}}$
- See whether J is less than/equal to 0.

Experimental setup

$$|\Psi\rangle = \frac{1}{\sqrt{1+r^2}}\left(|V\rangle_A |H\rangle_B + r\,|H\rangle_A |V\rangle_B\right)$$

- Eberhardt state used (note that Bell states ψ+ and ψ- correspond to r={1,-1})
- Why? Performance of the setup determines the optimal value of r.
- Values used: r=-2.9, a1 = 94.4◦ , a2 = 62.4◦ , b1 = −6.5◦ , and b2 = 25.5◦



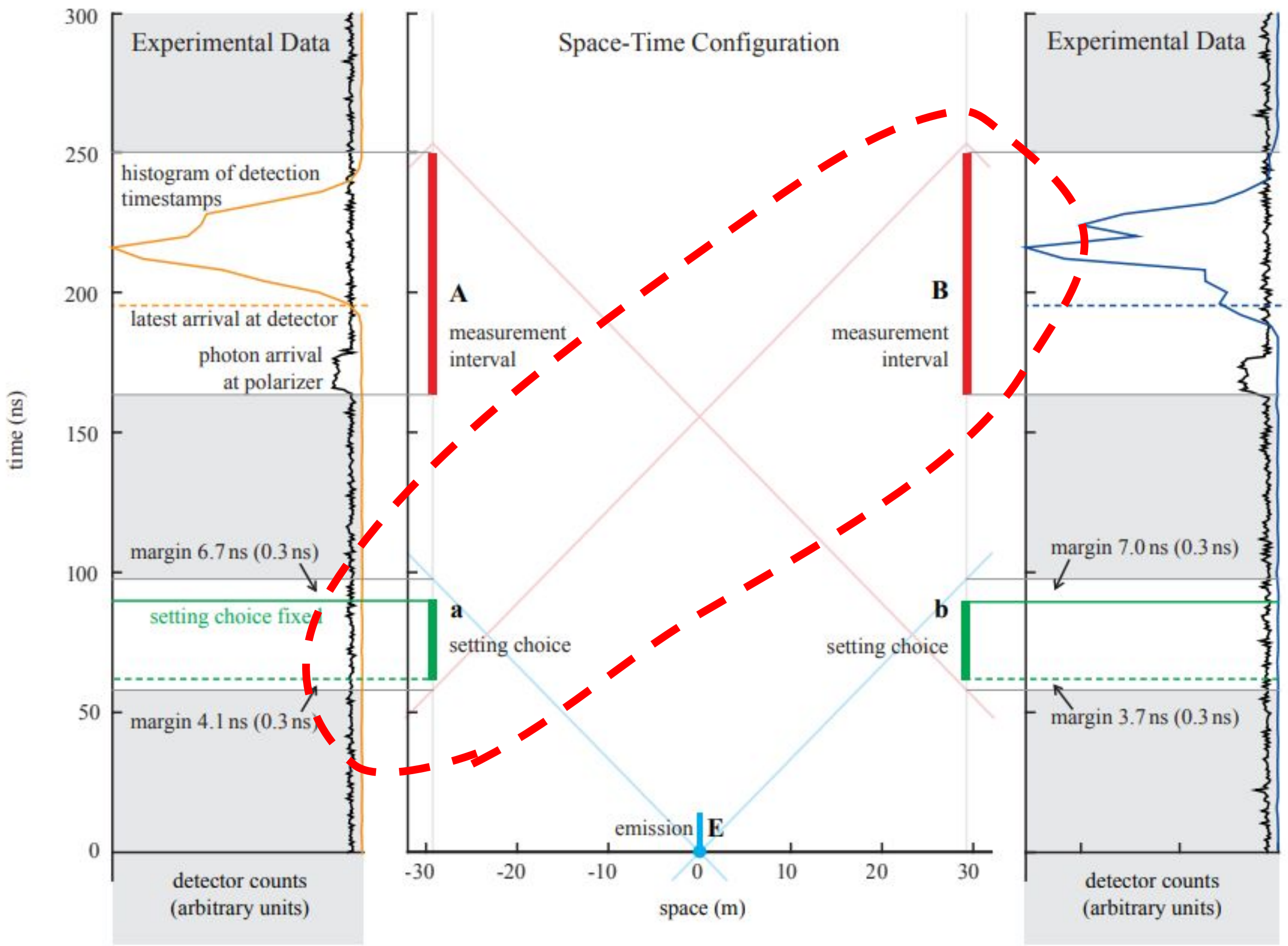
Locality and freedom of choice loopholes

- Locality:

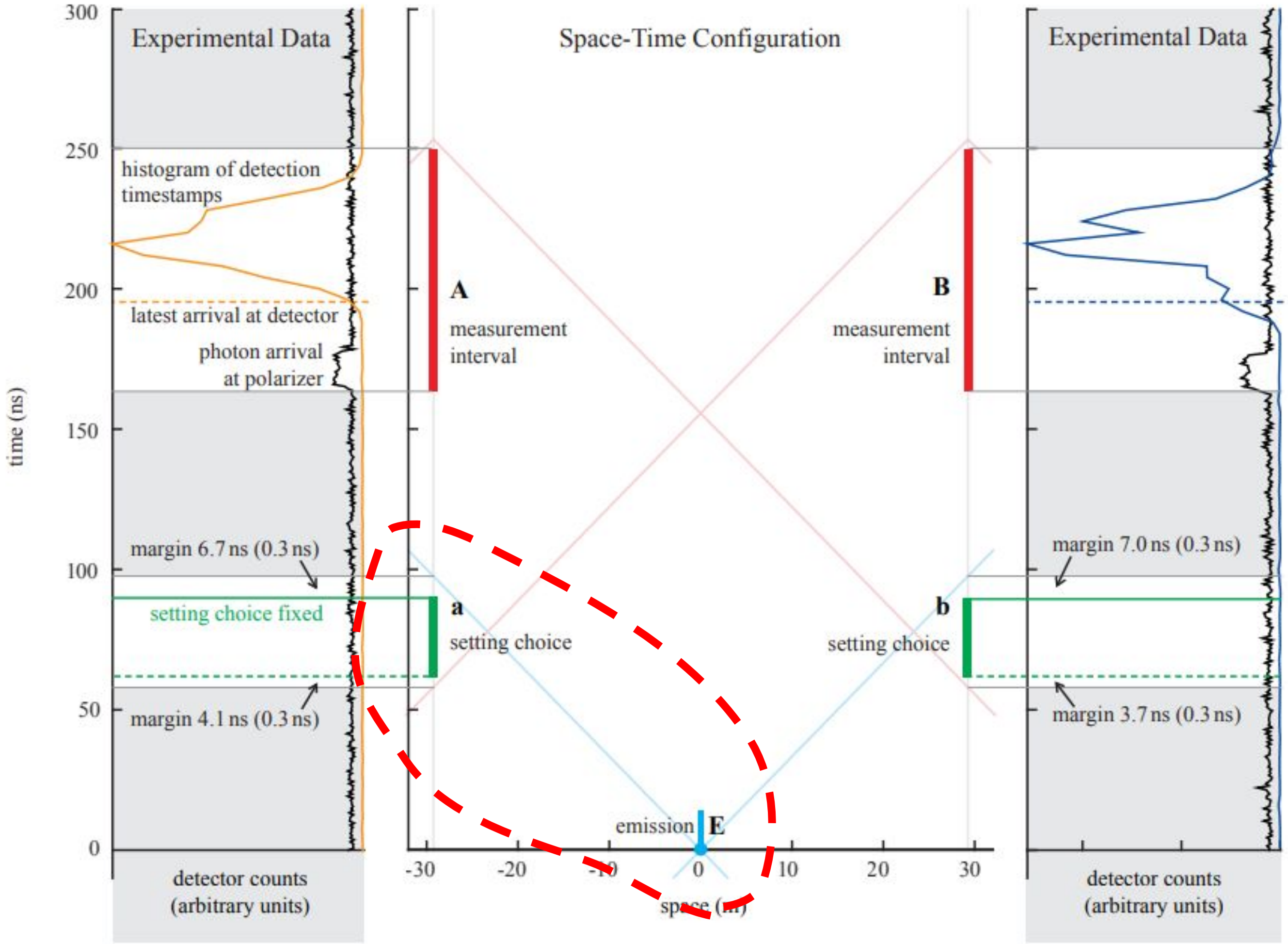
  Open if the setting choice or the measurement result of one side could be communicated to the other side in time to influence the measurement result there.

- Freedom of choice:

  Open if the settings are not generated independently at the measurement sites and spacelike separated from the creation of the particles measured (assuming λ originating with the particles to be measured).



Locality loophole



Freedom of choice loophole

Other loopholes?

- Memory loophole: past measurements can influence future measurements.

  Solved by: not assuming I.I.D data.

- Coincidence time loophole: if defined post-hoc, detection times can be influenced by hidden variables.

  Solved by: defining detection windows locally and before beginning of experiment.

Other loopholes?

- **Memory loophole: past measurements can influence future measurements.**

  Solved by: not assuming I.I.D data.

- Coincidence time loophole: if defined post-hoc, detection times can be influenced by hidden variables.

  Solved by: defining detection windows locally and before beginning of experiment.

Other loopholes?

- Memory loophole: past measurements can influence future measurements.

  Solved by: not assuming I.I.D data.

- **Coincidence time loophole: if defined post-hoc, detection times can be influenced by hidden variables.**

  Solved by: defining detection windows locally and before beginning of experiment.

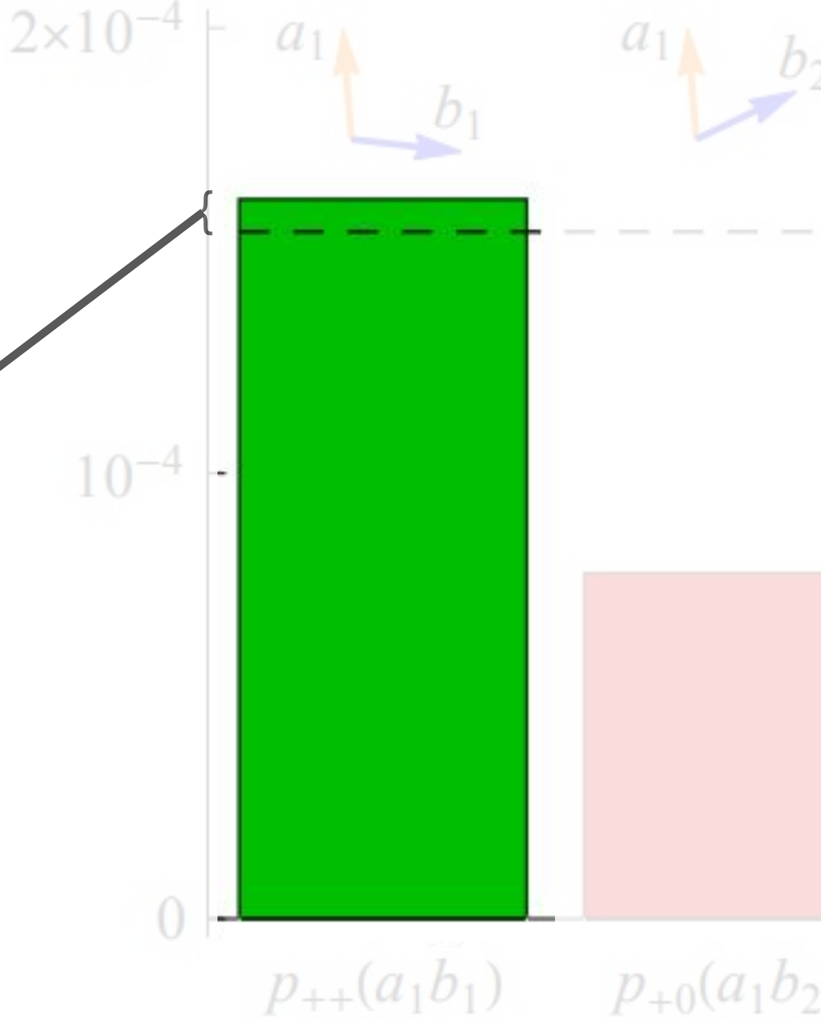
Results and conclusion

- J value
- Statistical significance
- Remaining loopholes?

Results and conclusion

- **J value (0.00004 theoretical)**
- Statistical significance
- Remaining loopholes?

$$0.00000727$$



Results and conclusion

- J value
- **Statistical significance**
- Remaining loopholes?

$$p = 3.74 \times 10^{-31} = 0.000000000000000000000$$

Results and conclusion

- J value
- **Statistical significance**
- Remaining loopholes?

$$p = 3.74 \times 10^{-31} = 0.00000000000000000000$$

(probably not the limiting factor in this study…)

Results and conclusion

- J value
- Statistical significance
- **Remaining loopholes?**

*"... we reduce the possible local-realist explanations to **truly exotic** hypotheses. [...] It has been suggested that setting choices determined by events from distant cosmological sources could push this limit back by billions of years."*